

Erdős Problem #400

EP-400

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

For any $k \geq 2$ let $g_k(n)$ denote the maximum value of $(a_1 + \cdots + a_k) - n$ where $a_1, \dots, a_k$ are integers such that $a_1! \cdots a_k! \mid n!$. Can one show that $\sum_{n \leq x} g_k(n) \sim c_k x \log x$ for some constant c_k ? Is it true that there is a constant c_k such that for almost all $n < x$ we have $g_k(n) = c_k \log x + o(\log x)$?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>